Data Mesh Standards Cascade

January 28, 2026

Agenda:

1. Data Mesh Standards
2. UAT Issues
3. GDM Compliance Report
4. Escalation Process/Onehub ticket

S3 Buckets

Bucket Naming Convention

1. **Root**
   - s3://<organization-name>-<environment>-<accountid>-<region>-root/...
   - s3://<organization-name>-<environment>-<accountid>-<region>-<business_unit>-root/...

2. **Landing**
   - s3://<organization-name>-<environment>-<accountid>-<region>-<gdm>-<business_unit>-landing/<source_name>/***raw***/date=<YYYYMMDD>
   - s3://<organization-name>-<environment>-<accountid>-<region>-<gdm>-<business_unit>-landing/<source_name>/***in process***/date=<YYYYMMDD>
   - s3://<organization-name>-<environment>-<accountid>-<region>-<gdm>-<business_unit>-landing/<source_name>/***archive/done***/date=<YYYYMMDD>
   - s3://<organization-name>-<environment>-<accountid>-<region>-<gdm>-<business_unit>-landing/<source_name>/***archive/failed***/date=<YYYYMMDD>

3. **Per Workspace / Business Unit / Line of Business**
   - (Parent Bucket) s3://<organization-name>-<environment>-<accountid>-<region>-<gdm>-<workspace>/...

Different structure for DMS Ingestion

1. The behavior of AWS DMS tasks in storing data uses the following output location:

   s3://bucket-name/<root path>/*database_schema_name/table_name/<actual-files>*.

   DMS does **not** allow modification in the sequence of the database schema, table and actual files.

2. Due to the restriction above, a different structure was implemented for DMS, as follows:

   s3://bucket-name/<root path>/**<status>**/**<databases>**/**<processing type>**/<**source IP address**>/*<database_schema_name>/<table_name>/<actual-files>*

   - **Status** pertains to *raw*, *in process*, *archive done*, and *archive failed*;
   - **Databases** pertaining to the source type if it's a *database* or *non-database*; and
   - **Processing type** pertaining to *full* or *incremental* table extract.

Workspaces

No.	Workspace Type	Description
1	NONPROD	Used for design, development, and experimentation of data pipelines, notebooks, and models.
2	PET	Acts as a validation and stabilization layer before production deployment.
3	PROD	Runs business-critical, production workloads with high reliability and security.

	NONPROD	PET	PROD	Sample
Workspace	<LOB>_NONPROD	<LOB>_PET	<LOB>_PROD	WDE_NONPROD, WDE_PET, WDE_PROD

Catalogs

Catalog Types

Each workspace should utilize the following catalog types for the purposes provided.

No.	Catalog Type	Description
1	Standard	This catalog contains the actual "day-to-day" data, which includes all the operational data used in regular business activities.
2	Metadata	This catalog houses metadata tables and attributes.
3	Exception	This catalog should only be used for specific use cases to which the catalog standards cannot be applied, such as: •Exploratory data analysis and discovery •Ad hoc or one-time data processing requests •High-priority requests or requests from top-level management

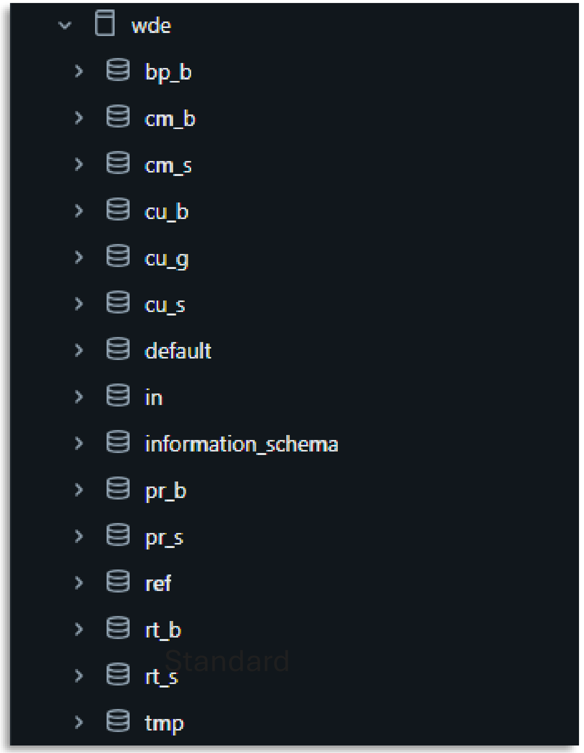
	DEV	PET	PROD	Sample
Standard Catalog	<LOB>_DEV	<LOB>_PET	<LOB>	WDE_DEV, WDE_PET, WDE
Metadata Catalog	<LOB>_LB_DEV_M	<LOB>_LB_PET_M	<LOB>_LB_M	WDE_LB_DEV_M, WDE_LB_PET_M, WDE_LB_M
Exception Catalog	<LOB>_DEV_E	<LOB>_PET_E	<LOB>_E	WDE_DEV_E, WDE_PET_E, WDE_E

Catalogs

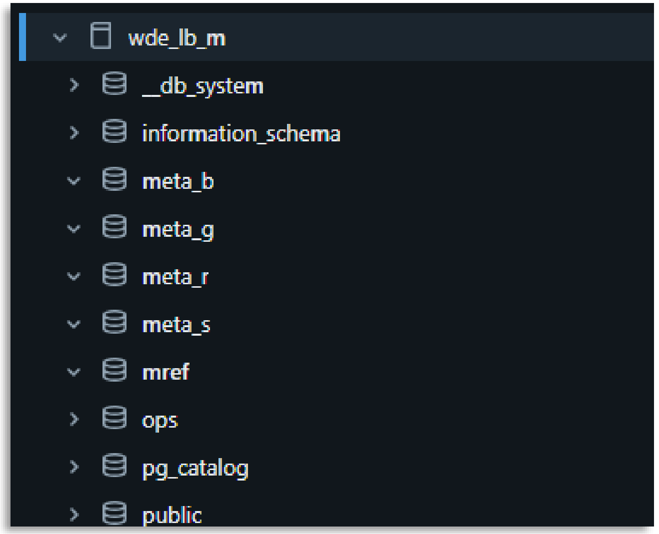
Exception Catalog Guidance

1. Use of the exception catalog should be strictly limited to the defined exception scenarios.
2. Furthermore, in case an exception imposes a significant risk on data security and/or data integrity, the exception must not be allowed even if it falls within scenarios.
3. Workspace Owners should be responsible for ensuring proper use of the exception catalog.
4. Since data in the exception catalog lacks sufficient governance, it will not be accepted for any data operations support.
5. Data in the exception catalog should not be shared outside the initial workspace. If data needs to be shared to other workspaces, the standard catalog must be utilized.

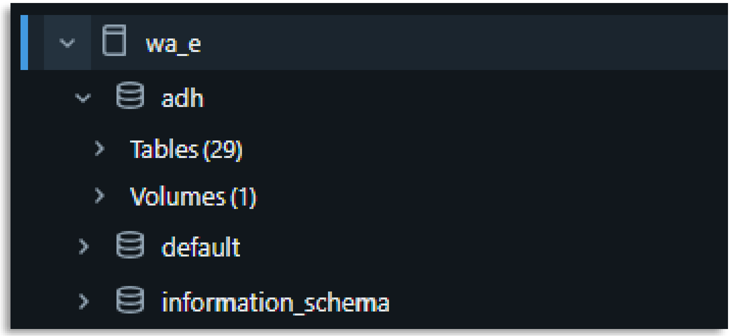
Catalogs



Standard



Metadata



Exception

Workspaces & Catalogs

PLDT Smart Group IT

Workspace & Catalog Codes

1. *GDM Naming Convention Codes - [link]*

#	Workspace group / LOB	Code
1	Wireless Data Engineering	WDE
2	Fixed Data Engineering & Visualization	FDE
3	Wireless Analytics	WA
4	Network Analytics	NWA
5	Data Governance & Quality	DGQ
6	Wireless Data Visualization	WDV
7	Data Science & AI	DS
8	Enterprise Analytics	ENA
9	Home Analytics	HA
10	Customer Experience	CX
11	Revenue Assurance & Fraud Mgt (2027)	RAFM
12	End users (including finance, sales, ..)	EUS*

Data Domain	Code
Customer	CU
Product	PR
Market & Sales	MS
Corporate or Enterprise	CR
Service	SE
Resource or Technology	RT
Business Partner	BP
Common	CM
Integration	IN

Workspaces & Catalogs

Workspace & Catalog Naming Convention

	DEV	PET	PROD	Sample
Workspace	<LOB>_NONPROD	<LOB>_PET	<LOB>_PROD	WDE_NONPROD, WDE_PET, WDE_PROD
Standard Catalog	<LOB>_DEV	<LOB>_PET	<LOB>	WDE_DEV, WDE_PET, WDE
Metadata Catalog	<LOB>_LB_DEV_M	<LOB>_LB_PET_M	<LOB>_LB_M	WDE_LB_DEV_M, WDE_LB_PET_M, WDE_LB_M
Exception Catalog	<LOB>_DEV_E	<LOB>_PET_E	<LOB>_E	WDE_DEV_E, WDE_PET_E, WDE_E

Schemas & Tables

Schema & Table Naming Convention

	BRONZE	SILVER	GOLD	Sample
Schema	<DATA DOMAIN CODE>_B	<DATA DOMAIN CODE>_S	<DATA DOMAIN CODE>_G	CU_B CU_G
Schema – Metadata	META_B	META_S	META_G	META_B, META_G
Schema – Reference	REF *(Should only contain manual and derived references, mapping, whitelist, report filters)*			REF.OFSC_MATCODE_TRANS
Schema – Metadata Reference	MREF *(Schema for metadata reference tables)*			MREF.JOB_DEPENDENCY, MREF.DQ_RULES_MASTER
Table	<SOURCE CODE>_ <FREQUENCY OF DATA UPDATE>_ < SOURCE TABLE NAME>	<DATA BUSINESS GROUP>_<DATASET NAME>	<F- FACT OR D- DIMENSION>_<USE CASE/REPORT NAME/PROJECT NAME>_ <GRANULARITY > *Note: Granularity is only applicable for Fact table*	OCS _DLY_SUBSTATE EN_OSMRDETAILS F_CX_CHATBOT_30M D_CX_CHANNEL
Reference Table	<SOURCE NAME (if applicable)> _ <TABLE/COLUMN/DATA ELEMENT NAME> _ <USE CASE>_GLOBAL (if common reference table)			REF.OFSC_MATCODE_TRANS REF.CALENDAR_GLOBAL

Standard Catalog – Schemas & Tables

(Standard) schemas and tables contain the actual business and/or operational data (i.e., customer, network, product, etc.)

Schema / Table	NAMING CONVENTION		
	BRONZE	SILVER	GOLD
(Standard) Schema	<DATA DOMAIN CODE>_B CU_B	<DATA DOMAIN CODE>_S CU_S	<DATA DOMAIN CODE>_G CU_G
(Standard) Table	<SOURCE CODE>_ <FREQUENCY OF DATA UPDATE>_ < SOURCE TABLE NAME>	<DATA BUSINESS GROUP>_<DATASET NAME>	<F- FACT OR D- DIMENSION>_<USE CASE/REPORT NAME/PROJECT NAME>_ <GRANULARITY >
Samples	CU_B.AIA_DLY_HOME_SUBS_PROFILE	CU_S.WIWE_SWITCH_SUBS_USAGE_DLY	CU_G.F_TMT_FIXED_SUBSBASE_MLY CU_G.D_CX_CHANNEL

- Given the broad nature of silver tables, other tagging such as <SOURCE CODE> or <GRANULARITY> may be added within the <DATASET NAME> **if the data set cannot be described without these tagging.**
- Indicator for frequency of data update and granularity uses the same list of values, the interpretation **depends on the type of table**, i.e. bronze or gold.
- For **table views**, add **'vw'** at the beginning of the table name.

PLDT Smart
Group IT

Standard Catalog – Schemas & Tables

Reference schemas and tables contain the relevant business reference data complementing the standard schemas and tables.

Schema / Table	NAMING CONVENTION
Reference Schema	REF
Reference Table	<SOURCE NAME (if applicable)> _ <TABLE/COLUMN/DATA ELEMENT NAME> _ <USE CASE>_GLOBAL (if common reference table)
Samples	REF.PRODUCT_WALLET_GLOBAL REF.SDR_PRODUCTID

- The reference schema is used regardless of medallion.
- **Global reference tables** contain reference data for data elements or reference objects that should be uniform across different data domains. These are mastered for consumption by multiple workspaces to improve data consistency. Examples of these are brand, location, calendar, etc.

Standard Catalog – Schemas & Tables

Temporary schemas (and tables) are used to create aggregated tables in the silver schemas.

Schema / Table	NAMING CONVENTION
Temporary Schema	TMP
Temporary Table	TMP_<TABLE NAME>
Samples	TMP_BRAND_REFERENCE_PNL

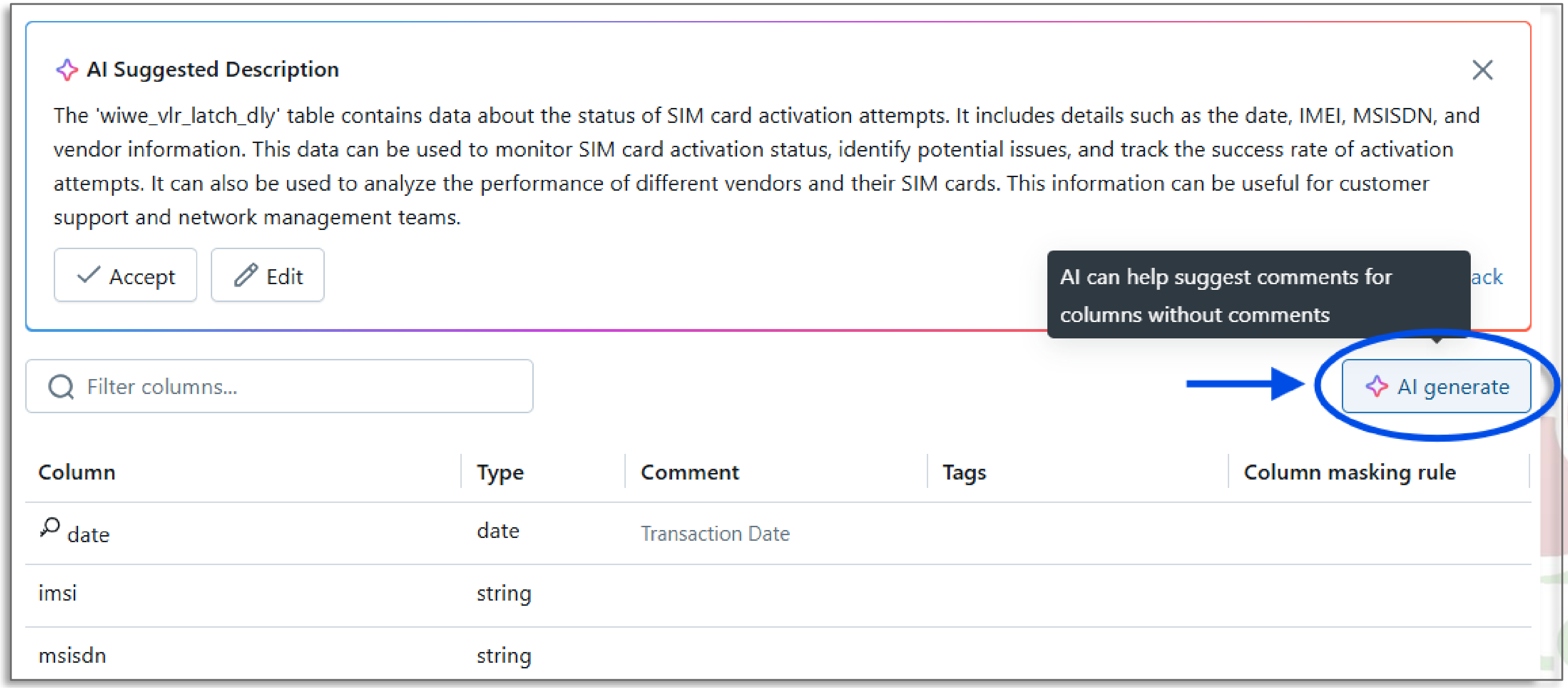
- Temporary tables must be marked with “tmp” at the beginning of the table name.
- Temporary tables must be stored in the temporary schema.
- Temporary tables must be truncated before and after running the job.
- Data quality implementation is not required on temporary tables.

Metadata Catalog – Schemas & Tables

Schema	NAMING CONVENTION	Tables
Metadata on standard tables	META_B META_S META_G	B/S/G_REF_EVENT_LOGS B/S/G_REF_PIPELINE_FILE_METADATA B/S/G_REF_PIPELINE_FILE_TO_DQ_TRANSACTION B/S/G_REF_PIPELINE_METADATA B/S/G_REF_PIPELINE_TO_DQ_TRANSACTION
Metadata on business reference tables	META_R	R_REF_EVENT_LOGS R_REF_PIPELINE_FILE_METADATA R_REF_PIPELINE_FILE_TO_DQ_TRANSACTION R_REF_PIPELINE_METADATA R_REF_PIPELINE_TO_DQ_TRANSACTION
Metadata on data processing tables	MREF	JOB_DEPENDENCY PIPELINE_CONFIGURATIONS REF_DQ_RULES_MASTER REF_PIPELINE_TO_RULE_MAPPING RETENTION_MASTER_TABLE S3_MANIFEST_DBX_JOB_MAPPING S3_MANIFEST_DBX_JOB_MAPPING_TABLE TABLE_DEPENDENCY
Metadata for data operations tables (job-related system tables)	OPS	JOB_RUN_TIMELINE JOB_TASK_RUN_TIMELINE JOB_TASKS JOBS

Table & Column Comments (Data Definitions)

1. Table and column descriptions must be available in the GDM. These definitions may be created with the help of AI, using AI Generate function.
2. Definitions must be generated during development and validated up until testing. Developers and engineers must ensure that definitions are in sync across environments.
3. Data definitions should align with the business context in which the tables are used.

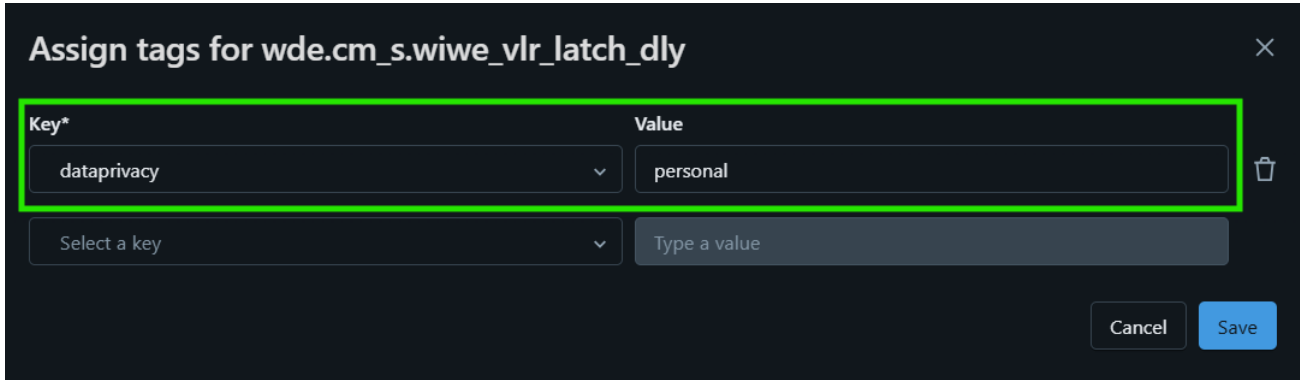


PLDT Smart Group IT

Data Classification via Tags

1. If the table contains personal information or sensitive personal information, tag the table as follows:
   Key: ***dataprivacy***, Value: *personal*
   Key: ***dataprivacy***, Value: *sensitive*

2. If the table contains restricted data, tag the table as follows:
   Key: ***infoclass***, Value: *restricted*



Note: All data in GDM should be considered as **confidential**, therefore apply tag only for the higher classification such as restricted data.

Data Classification via Tags

Definitions as provided by DPO & CSOG

Policy on the Protection & Use of Personal Data - link

3.11.	Personal Data	refers to all types of Personal Information, Sensitive Personal Information, and Privileged Information.
3.13.	Personal Information	refers to any information, whether recorded in a material form or not, from which the identity of an individual is apparent or can be reasonably and directly ascertained by the entity holding the information, or when put together with other information, would directly and certainly identify an individual.
3.21.	Sensitive Personal Information or SPI	refers to personal information: (1) About an individual's race, ethnic origin, marital status, age, color, and religious, philosophical, or political affiliations; (2) About an individual's health, education, genetic or sexual life of a person, or to any proceeding for any offense committed or alleged to have been committed by such person, the disposal of such proceedings, or the sentence of any court in such proceedings; (3) Issued by government agencies peculiar to an individual which includes, but not limited to, social security numbers, previous or current health records, licenses or its denials, suspension or revocation, and tax returns; and (4) Specifically established by an executive order or an act of Congress to be kept classified.

Information Asset Management Standards - link

5.2. Classification Levels

5.2.1. Information assets are categorized into four classifications based on confidentiality and criticality:

Classification	Public	Internal	Confidential	Restricted
Confidentiality	Approved by PLDT Group for public consumption	Approved for release to all employees	Accessible to authorized groups or individuals only.	
Criticality	Unauthorized disclosure or loss does not result to operational, reputational, legal, regulatory or financial impact to PLDT Group	Unauthorized disclosure or loss may cause inconvenience to business operations but does not result to financial, legal, regulatory or reputational impact to PLDT Group	Unauthorized disclosure or loss may cause operational, reputational, legal, regulatory or financial impact to PLDT Group, such as loss of customer or shareholder confidence, drop in stock value, or potential violation of laws and regulations	Unauthorized disclosure or loss will certainly cause operational, legal, regulatory, reputational or financial damage to PLDT Group, such as disruption of operations, loss of market share, severe drop in stock value, or violation of laws and regulations
Examples	• Released financial statements • Press statements	• Company policies and standards • Internal memos and advisories	• Customer records • Employee records • Network diagrams	• Impending mergers or acquisitions • Unreleased technologies • Passwords

Data Retention

Data Retention Periods – AWS (landing zone) & Databricks (medallion/catalog)

Layer	Duration in days (All except FDE)	Duration in days (FDE)
AWS S3 Landing Zone	365	3650
Bronze	365	3650
Silver	1095	3650
Gold	1825	3650
Exception catalog	92	92

- Retention periods are not applicable to metadata and reference tables.
- The retention periods above should be implemented by all workspaces within the GDM, except for the S3 landing zone, bronze, silver and gold layer of the FDE workspace (for which 10 years or 3650 days retention has been defined)
- In case of other exceptions, for instance for analytics use cases within the exception catalog, this should be raised to the Workspace Owners for justification and approval.

Data Security – Access Management

PLDT Smart
Group IT

The following requirements must be applied.

1. **Role-based Access Control (RBAC)**. RBAC means assigning access permissions based on the roles assigned to users within an organization, ensuring that individuals only access the resources necessary for their job functions.

2. **Principle of Least Privilege (PoLP).** PoLP means individuals or systems have the bare minimum permissions required to execute their specific roles or tasks.

3. **DBX Access Matrix for DGQ Access Group**

PRIVILEGE	DEV	PET	PROD
Use Catalog	YES	YES	YES
Use Schema	YES	YES	YES
Browse	YES	YES	YES
Select	YES	YES	YES
Add Tag	YES	YES	NO
Modify	YES	YES	YES*

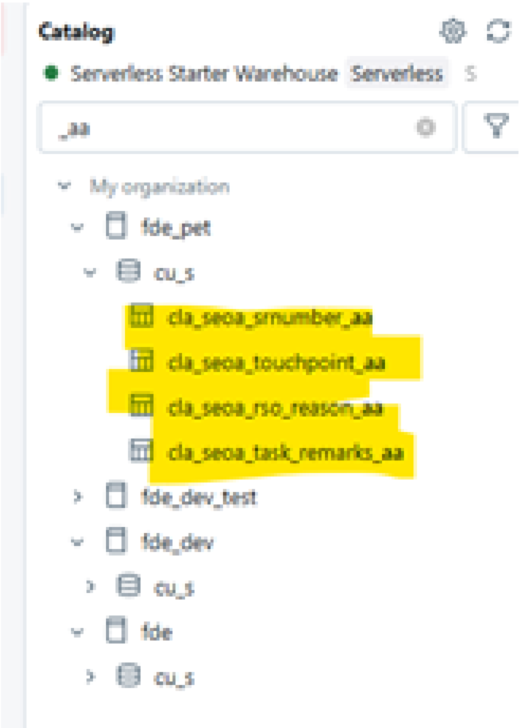
- Applied to GSG SMART GDM Databricks Data Quality Assurance DGQ *(PLDT no in use as all DGQ users are Smart badges)*
- Applied at catalog level except for MODIFY in PROD
- (*) Modify in PROD should be for metadata catalogs only, needed by DQ to update DQ metadata tables

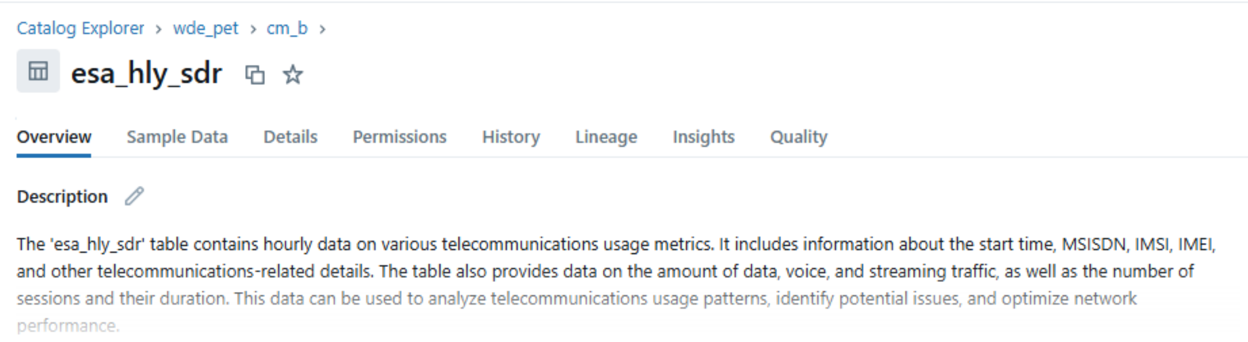
alapaap

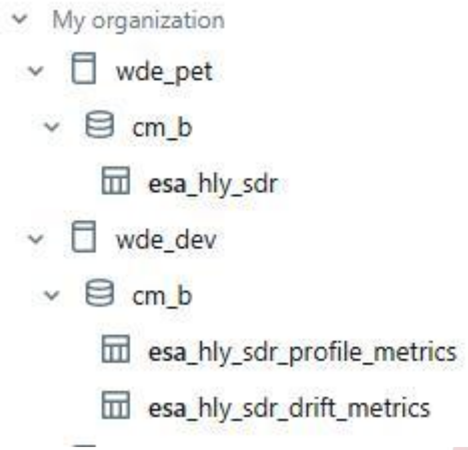
Common Data Governance Issues (UAT)

1. Structure and Naming Conventions

It was noted that certain tables did not comply with the established structure standards.





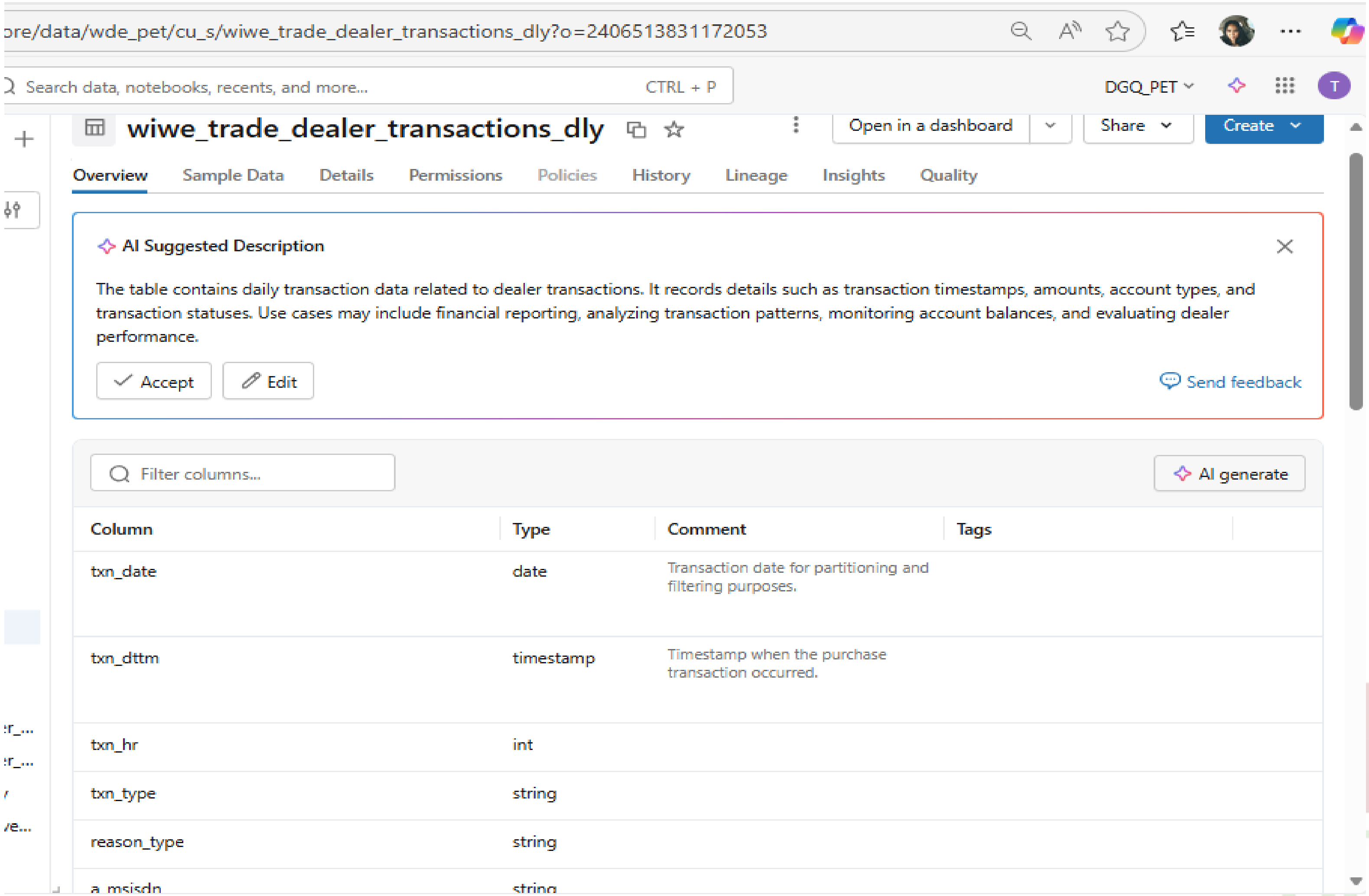


Data source and business group identifiers were often interchanged.

Incorrect codes were used for data sources.

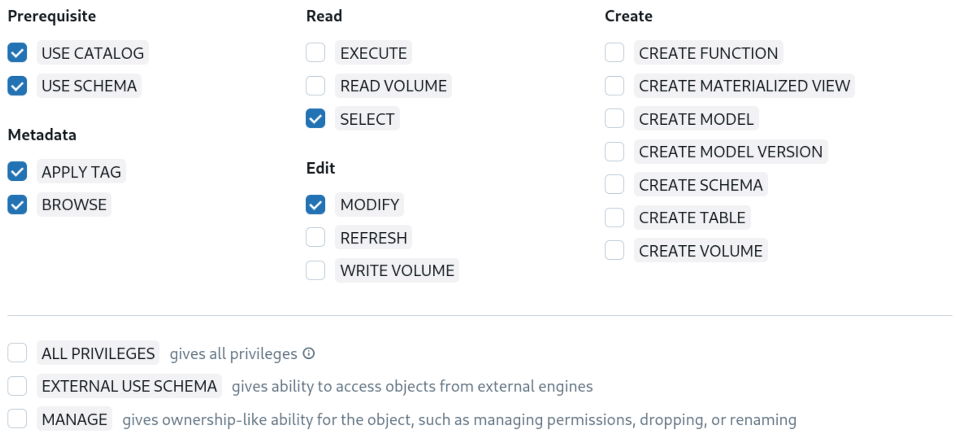
2. Table and Column Definition

Missing or incomplete data definitions.



3. Access Control and Permissions

Often required modifying permissions before the appropriate data became accessible.
Test users might have had access to sensitive production data or lacked access to required datasets.

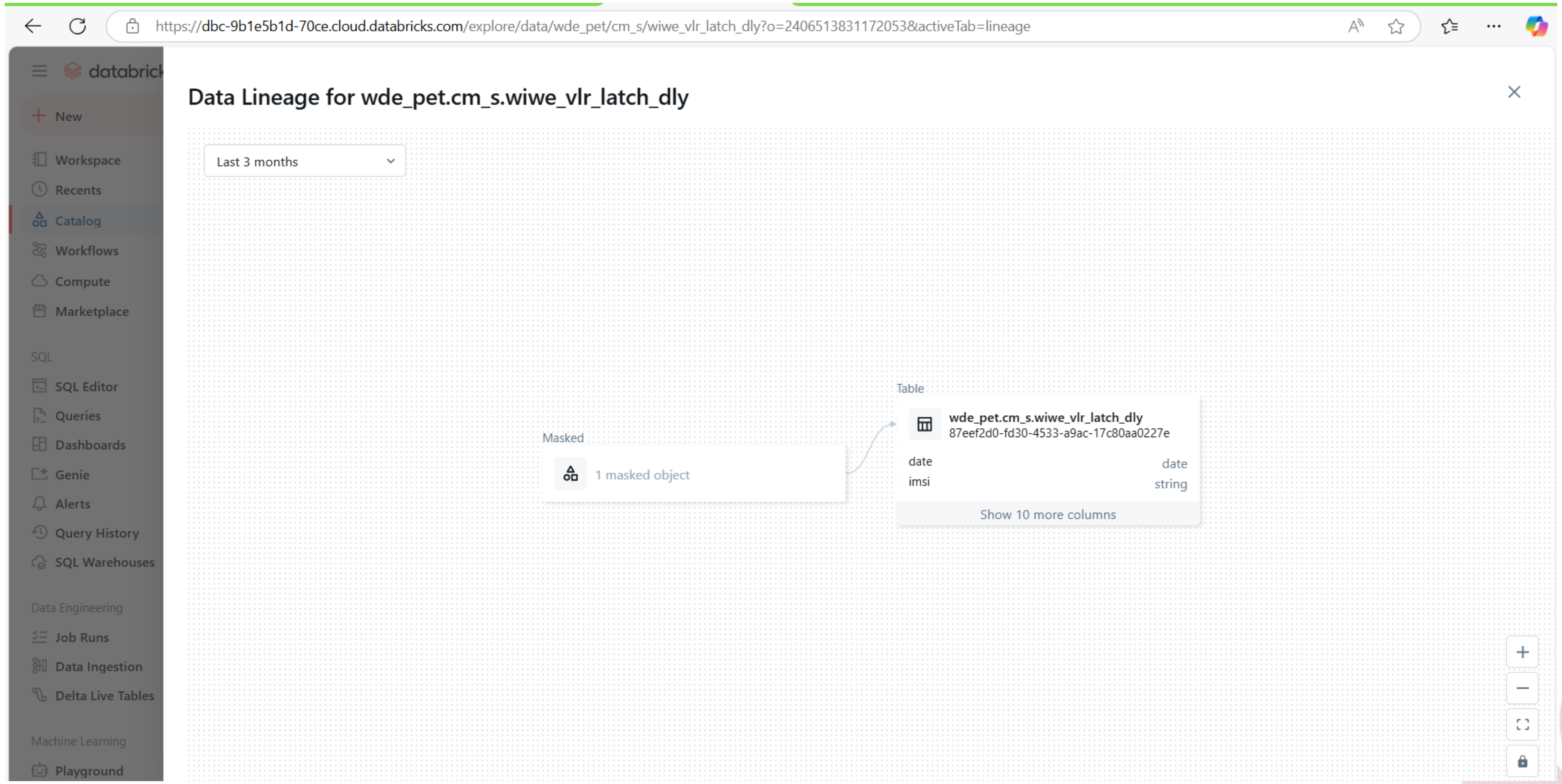


	DEV	PET	PROD
Use Catalog	yes	yes	yes
Use Schema	yes	yes	yes
Browse	yes	yes	yes
Select	yes	yes	yes
Add Tag	yes	yes	no
Modify	yes	yes	yes*

- Applied to GSG SMART GDM Databricks Data Quality Assurance DGQ
- Applied at catalog level except MODIFY in PROD
- (*) Modify in PROD for **metadata catalogs only**, this is needed by DQ to update DQ metadata tables

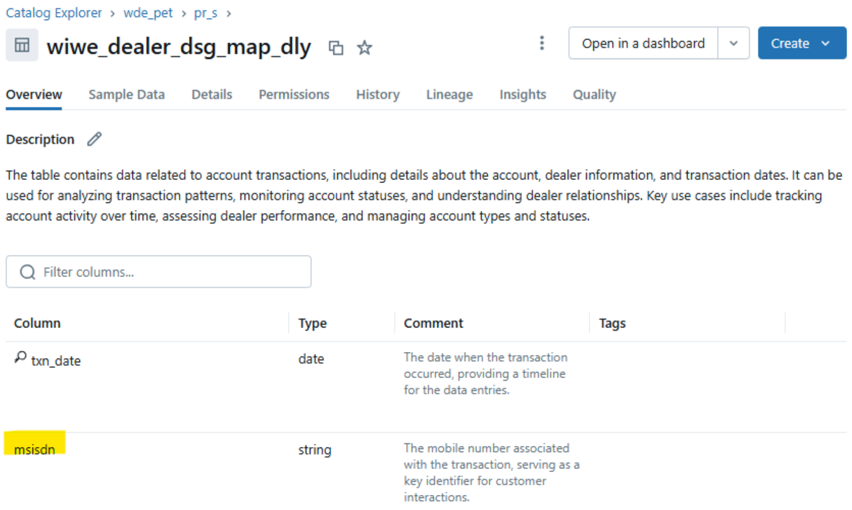
During the initial phases, permissions had to be manually adjusted before DG TEAM could access the necessary data.

4. Data Lineage Tracking

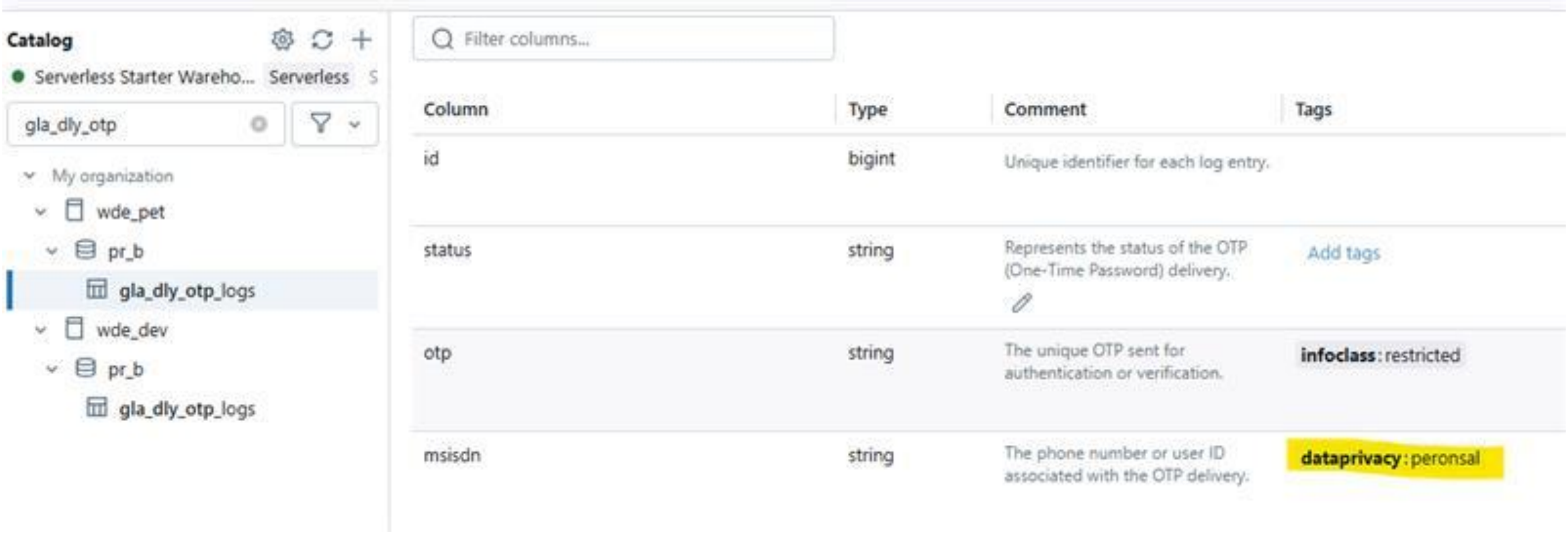


Tracking tool did not consistently capture all transformation steps, some objects were masked or skipped, resulting in incomplete visibility into data flow.

5. Data Classification and Tagging



Recurring issues on inconsistencies in data tagging. Critical tags like PII were sometimes missing or applied inconsistently across datasets.



Tag Name	Tag Value
dataprivacy	sensistive
dataprivacy	sensistive

6. Data Lifecycle Management

Table

	table_catalog	table_schema	table_name	table_owner	created	created_by
1	wde_pet	rt_b	ensa_hly_sdr_election_test	87eef2d0-fd30-4533-a9ac-17c80aa0227e	2025-04-02T15:15:26.083+08:00	87eef2d0-fd30-4533-a9ac-17c80aa0227e

There was no proper cleanup of test artifacts, leaving behind unused/test tables that cluttered the workspace and increased potential risks.

GDM Compliance Report

January 28, 2026

GDM Compliance Report

Background

Following our migration to the cloud and adoption of Databricks within Data Mesh Architecture, the Data Governance Team has proposed implementing BAU monitoring. This initiative is intended to ensure ongoing oversight and stability across domains. To effectively manage risks, we recommend routine checks on:

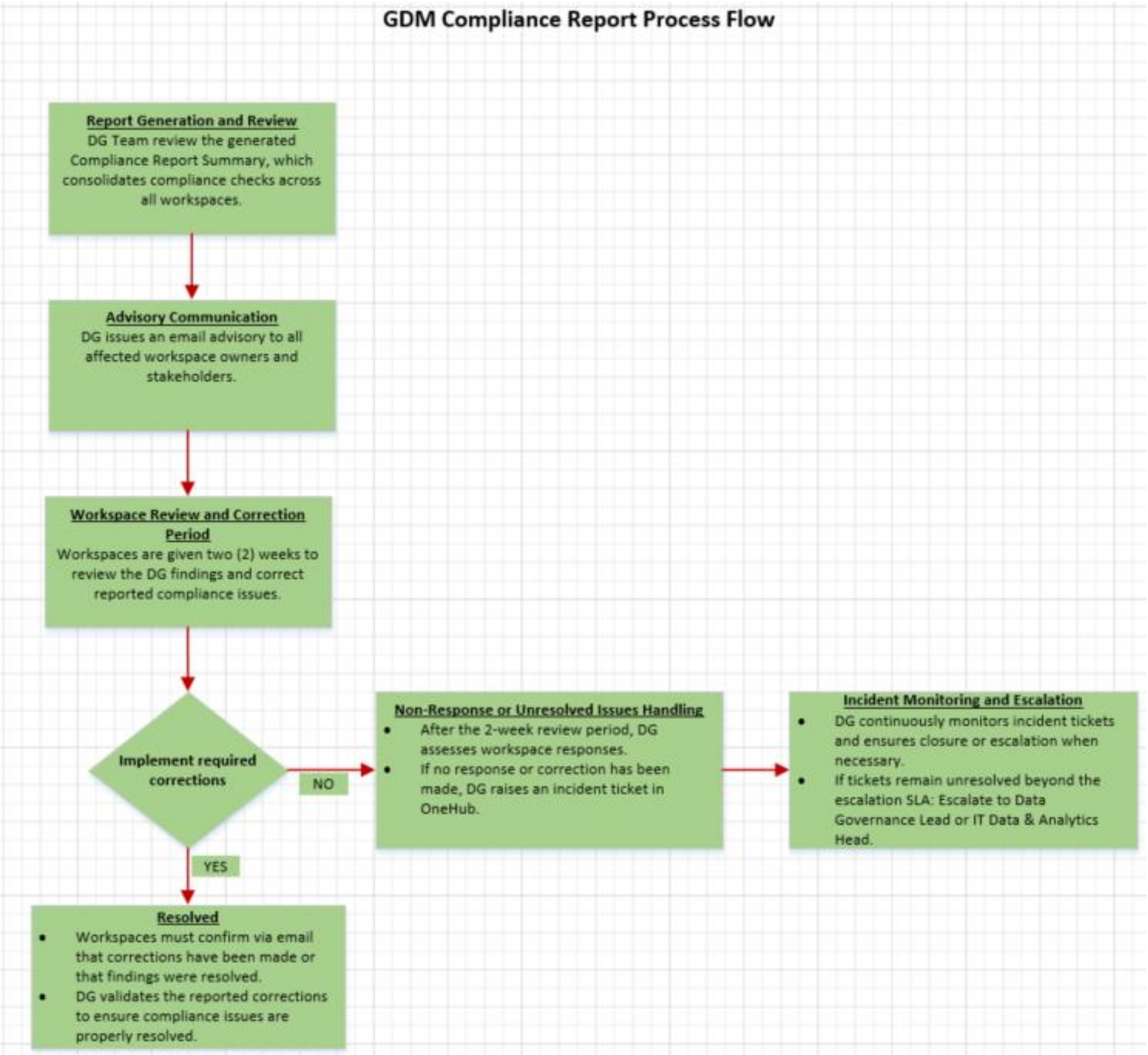
1. **Table Description** – Ensuring business and technical context is properly documented.
2. **Column Comment** – Enhancing business and technical clarity for data consumers.
3. **Naming Convention** – Promoting consistency and discoverability.
4. **Data Classification** – Supporting data sensitivity and compliance requirements.
5. **Ownership –** Ensuring accountability, traceability, and proper access governance.

Dashboard link: *DG Compliance Report*
POC: IT.Data.Governance@smart.com.ph

Metrics

Below are the metrics used for weekly compliance tracking:

Checks	Rule	KPI
Table description	Standard tables in Unity Catalog must include a description. The description should be complete, clear, and accurately reflect the table's purpose and usage. Existing legacy descriptions are acceptable.	% of tables with description. Formula: (Tables with valid description / Total tables) * 100
Column comment	Each column in the standard tables in Unity Catalog must have a description. The description should be clear, complete, and accurately convey its purpose and usage. Existing legacy descriptions are acceptable.	% of columns with comments. Formula: (Columns with valid comment / Total columns) * 100
Column Duplicate	Checks consistency of column comments across tables. Each column name should have one standard comment (Many-to-One) if it serves the same purpose and usage across datasets. Inconsistent comments indicate documentation misalignment. (e.g. dbx_process_dttm should have the same meaning and comment across all tables where it appears.)	The expectation is "Many-to-One" — multiple occurrences of the same column name should map to one distinct comment. % of Pass (Pass / total records) * 100
Naming Convention	Table names follow the standardized structure and naming guidelines.	% of tables compliant with naming convention. Formula: (Tables matching structure and naming convention & regex / Total tables) * 100
Data Classification	Any column with PII, SPI, and Restricted pattern (msisdn, lacci, etc.) must have tag. This check will rely on the initial tagging of developers/engineers to validate table to table consistency and accuracy.	% of columns correctly classified. Formula: Number of columns tagged correctly / Total number of tagged columns) * 100
Ownership	Tables must **not** be created or deployed using personal accounts.	% of tables created using approved non-personal accounts (Tables created by approved accounts / Total tables) × 100


GDM Compliance Report Process Flow
Report Generation and Review
DG Team review the generated Compliance Report Summary, which consolidates compliance checks across all workspaces.
Advisory Communication
DG issues an email advisory to all affected workspace owners and stakeholders.
Workspace Review and Correction Period
Workspaces are given two (2) weeks to review the DG findings and correct reported compliance issues.
Implement required corrections
NO
YES
Non-Response or Unresolved Issues Handling
After the 2-week review period, DG assesses workspace responses.
If no response or correction has been made, DG raises an incident ticket in OneHub.
Incident Monitoring and Escalation
DG continuously monitors incident tickets and ensures closure or escalation when necessary.
If tickets remain unresolved beyond the escalation SLA: Escalate to Data Governance Lead or IT Data & Analytics Head.
Resolved
Workspaces must confirm via email that corrections have been made or that findings were resolved.
DG validates the reported corrections to ensure compliance issues are properly resolved.